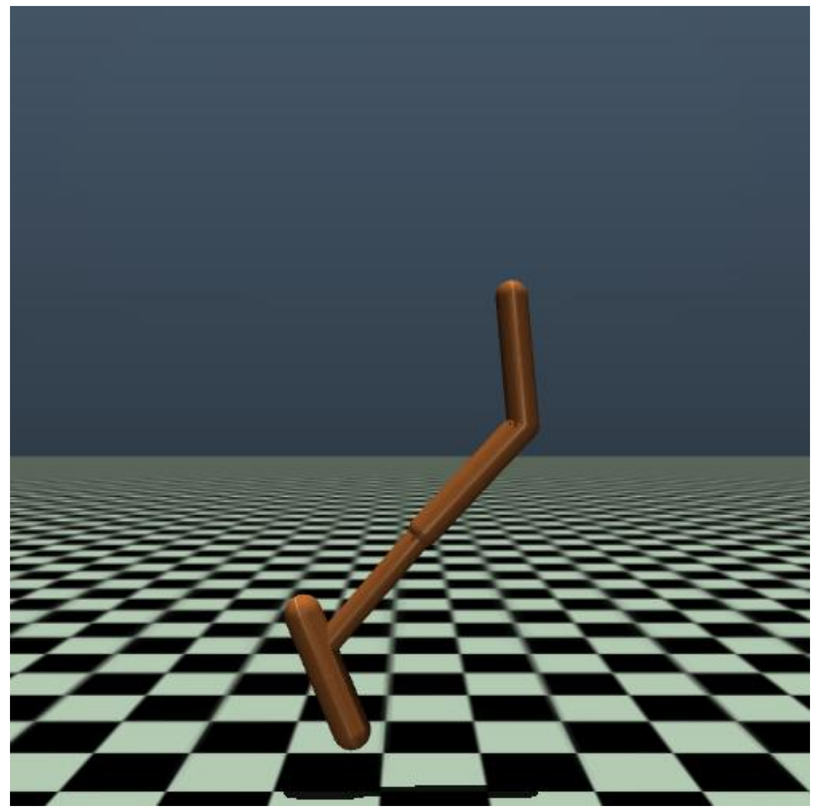
Problem: Parameterizing Policies

In Lecture 2, we learned that policy $\pi(a|s)$ is a conditional action distribution. To facilitate the learning, starting in Lecture 3, we consider parametric policies $\pi_\theta(a|s)$. You may refer to the slides **pp. 34-36** of Lecture 3 as a starting point when answering the questions below.

1) Discrete case: Suppose the state space $\mathcal{S}$ and action space $\mathcal{A}$ are both finite. How would you design the π_θ so that you can express all kinds of stationary policies?

2) Continuous case: Consider the classic RL task **Hopper**, which is a 3-DoF robot (https://gymnasium.farama.org/environments/mujoco/reacher/) with a continuous state space $\mathcal{S} = \mathbb{R}^{11}$ and an action space $\mathcal{A} = [-1,1]^3$. How would you design the π_θ so that you can express all kinds of stationary policies?



3) Constrained case: Based on 2), suppose due to some physical limitations, we need to enforce an additional action constraint as $\|a\|_2 \leq 1$. What changes would you make to ensure that the output actions are ***feasible*** (i.e., satisfies the constraint)?

For problem 1

Based on the lecture slides, for a discrete action space, we can use the softmax policy to model the policy: $\pi_\theta(a|s) = \dfrac{\exp(\theta_{s,a})}{\sum_{a' \in A} \exp(\theta_{s,a'})}$

This ensure that $\pi_\theta(a|s)$ is a valid probability distribation (all probabilities sum to 1)

Extensions:

Linear softmax policy

$$\pi_\theta(a|s) = \frac{\exp(\theta^T \phi_{s,a})}{\sum_{a' \in A} \exp(\theta^T \phi_{s,a'})}$$

where $\phi_{s,a}$ is a feature vector represent the state-action pair.

Neural softmax policy

$$\pi_\theta(a|s) = \frac{\exp(f_\theta(s,a))}{\sum_{a' \in A} \exp(f_\theta(s,a'))}$$ where $f_\theta(s,a)$ is a function modeled by a neural network.

For problem 2

In the Hopper task, the state space is $S = \mathbb{R}^{11}$ and the action space is $A = [-1,1]^3$.

Based on the lecture slides, for continuous action spaces, the most common approach is to use a Gaussian distribution: $a \sim N(f_\theta(s), \Sigma^2)$

This means that for a given state s, the action a is sampled from a normal distribution with mean $f_\theta(s)$ and variance σ^2

To derive $\pi_\theta(a|s)$ from Gaussian distribution

$$\pi_\theta(a|s) = \frac{1}{(2\pi)^{\frac{3}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(a-f_\theta(s))^T \Sigma^{-1}(a-f_\theta(s))\right)$$

For problem 3

To ensure the generated action a satisfies this constraint, we can apply the following modifications:

1. Normalization

First, sample an unconstrained action $\tilde{a} \sim N(f_\theta(s), \sigma^2)$, then normalize it:

$$a = \frac{\tilde{a}}{\max(1, \|\tilde{a}\|_2)}$$

If $\|\tilde{a}\|_2 \leq 1$, a remains unchanged. Otherwise, the a is scaled down to fit within the unit ball.

2. radial squashing (from paper)

apply tanh to the l_2 norm of the input vector, the scale the vector propotionally. This ensures that the action always remains with in the unit ball ($\|a\|_2 \leq 1$)